1.08 Land Spreading

1. **Purpose**

   The purpose of the Standard Operating Procedure is to identify the hazards and define the steps in land spreading with a manual and automatic transmission .

2. **Scope**

   This procedure is intended for use by the Silverpoint employees .

3. **Hazard Identification**

   - Physical Stress
   - Mental Stress
   - Personal Injury
   - Fatigue
   - Equipment Damage
   - Hidden Hazards
   - Deep Snow
   - Rocks
   - Potential Energy
   - Ergonomics
   - Equipment Failure /Stress
   - Release to the Environment
   - Vehicles , Mud , Ice , Animals , People , Weather
   - Slips , Trips , Falls , Lifting , Twisting , Pushing , Pulling
   - Soft Fields
   - Animal Holes

4. **EHS Requirements**

   - Fire Retardant Coveralls .
   - Steel Toed Boots .

- Safety Gloves.
- Hard Hat
- Hearing Protection

5. **References**

- Silverpoint Corporate Codes of Practice
- Alberta OH&S
- Silverpoint Task List
- Silverpoint THA 1.08

NOTE: Some Environmental Technicians we work with use Survey Stakes/Pylons or in some cases once you are shown the Water Body/Boundary areas there are no Markers used at all. It will be up to each Operator to determine what system Technicians are using to mark the Spread Fields and Water Bodies. Get a Map from the Tech and be sure to drive around the Spread Field to check size, Hazards etc.

6. **Procedure**

Enter the spread field; keep in mind any special instructions from the technician. When the field is soft you will have to use Tire Boss, refer to 2.04 Tire Boss Procedure. Check field conditions for hazards and obstacles as you enter. Also check the wind direction and try to spread into the wind. Get the truck into position for the new spread. If there are already existing spreads make sure to leave some space between them. Remember: **SPREADS CANNOT OVERLAP.**

Be sure to remove the trailer Hitch before raising the Tank or you will break off the 6 to 4 reducer.

6.1.1. Spread Rates:

The following is taken from the ERCB (Energy Resources Conservation Board) Directive 50.

To minimize migration and pooling of drilling wastes:

6.1.1.1. Summer operations: (non-frozen ground) are limited to a maximum LWD rate of 40 m3ha and spraying cannot occur within 100 m of any water body. **Silverpoint extends this to 140 meters.**

6.1.1.2. Winter operations: (frozen ground) are limited to a maximum LWD rate of 20 m3ha and spraying cannot occur within 200 m of any water body. **Silverpoint extends this to 250 meters.**

6.1.1.3. Do not spread on any land that is greater that a 5% grade, discuss with tech if in doubt about the grade of the field.

6.1.1.4. Avoid any rutting of the fields, any pooling of fluid, shale piles or overlapping of spreads, Fence damage, Garbage, Spills and shale piles must be cleaned up. Report it to the Tech. spread into the wind if possible or side wind

6.1.2. You may not be able to spread as far as the tech wants you to because its so rough, tell them you need to slow down or you will damage the truck, contact operations if you are having problems with the tech cooperating with you to find a smoother spread field.

6.1.3. When you are on your way to the field, if it starts to rain or has been raining, and you are second guessing about whether you should go into the field, **STOP** and phone your field tech or supervisor. **It is better to get a second opinion then to go into the field and damage it.**

6.1.4. The Tech you are working with should have shown you your spread field and put up the appropriate flagging and markers to aid you in identifying hazards.

Winter reminders:

- Tire chains are an aid in traction, not to allow entry to areas a vacuum truck would not normally drive in.
- Understand how to install them properly (SOP 2.03), understand the importance of keeping them tight, loose chains can damage TPC lines, and be sure they are the right size to fit your trucks tires. Understand the increased traction they provide and the effects on the truck drive train when power is applied.
- Avoid areas where snow is deep enough to conceal hidden hazards.
- Try to evaluate spread field during daylight hours on a regular basis, giving attention to overall landscape.
- If snow is deep enough to conceal hazards area should be avoided until it can be inspected during the day or plowed.
- Chances of a hidden hazard (farm equipment, rocks, etc) in a non-cultivated field are far more likely than a cultivated field.
- All operators are expected to evaluate an area they are about to enter as if they did not have tire chains on the truck. If an area that you are about to drive into looks like the truck couldn't get thru without chains on, move to a different area that has less snow so you can see the hazards.

6.2. Preparing to spread with Manual and Automatic Transmissions:

6.2.1. **Manual** - Stop the vacuum truck and set the brakes, put the transmission in gear (This is to ensure the transmission is not turning when you engage the PTO), engage the main PTO and the agitator PTO take transmission out of gear and **SLOWLY** release clutch. Increase engine speed until it reaches the specific RPM for that unit. The specific rpm is posted on the dash of the truck.

6.2.2. **Automatic** - Stop the truck and put the transmission into neutral, engage the PTO (on the automatics, the agitator runs off the main PTO), engage the agitator (either with a switch in the cab, or outside lever). Increase the engine speed until it reaches the specified RPM required. The required RPM is posted on the dash or console of the truck. Once this is done you can switch the transfer case into low range (**make sure that you go to neutral then to low**) to prepare for spreading. This is for an automatic transmission. Ask the proper procedure for switching the transfer case, so it does not get damaged.

6.3. Starting the agitator now will allow the maximum amount of time to stir the tank, the amount of shale that is in the vacuum tank will determine how long you need to agitate the load. Very heavy loads of shale may need to be agitated for up to 10 minutes or more.

6.4. Exit the truck and remove **all the caps** from the back of the truck and ensure the 6 to 4 reducer is on the spread valve and attach the spread plate.

6.5. You can now burp the truck refer to (**SOP 1.05 Burping**)

6.6. Next switch the 4-way valve over to pressure. You may now re-enter the truck.

6.7. Engage the Hybon and build pressure in the system until you can hear the pressure relief valves open. Engage the clutch and place the transmission into the appropriate gear for your spread rate. (Any questions regarding spread rates should be directed at your technician). You must be moving forward and have pressure in the tank before opening spread valve to prevent pooling.

6.8 When ever possible, spread into the wind or cross wind. Never spread with the wind as this maneuver will result in a truck covered in mud.

6.8.1. Automatic: Engage the Hibon and build up pressure in the system until you hear the pressure relief valves open. Release the brake and pick the appropriate gear to spread ratio.

6.9. **Automatic:** Push on the accelerator to move forward until you have reached proper RPM.

6.10. **Manual:** Slowly release the clutch and start moving forward until you reach the proper RPM.

6.10.1. At this point the Hibon should be on, and you should be at proper ground speed now open the 6" valve. Check the mirrors to ensure that you are spreading and that the spread is coming out of the spread plate evenly. If you notice the spread is not even, **close the spread valve and turn off Hibon, maintain speed until fully closed, and then stop the truck**. Then place the transmission in neutral and apply the brakes. Relieve the pressure in the tank with the 4-way valve. Check the spread plate for blockage and repeat 1.05 Burping the Vacuum Truck Procedure, if necessary.

6.11. Continue spreading with tank down until you have ½ of the load off the truck and then raise the tank no higher than the first stage. This will allow the remaining fluid to run to the back of the tank. **Make sure you do not raise the tank on a side hill or turn with tank raised. The Hoist on these trucks were never designed to be raised while the tank is full of fluid, the manufacture makes this very clear in the owner's manual.**

6.12. Continue spreading until the tank is empty. Shut off the Hibon pump and close the spread valve. You can now lower the tank and stop the truck. Place the transmission in neutral, set the park brake and exit the truck and walk around the unit scanning the ground for tripping hazards and rocks or other objects that could cause damage to the unit when you go to exit the spread field. Drain the cyclone, silencers and final filter then slowly relieve pressure from the system. Place 4-way valve on Vacuum. Make sure all caps are removed from other valves.

6.13. Engage Hibon on vacuum, build vacuum and open the spread valve to clear mud from it and then close it. Open the agitator valve to clear mud from agitator and valve then close it. Then open load valve to clear mud from it

6.13.1. Under special circumstances the agitator can be blown out using pressure, this is only used if you suspect there is debris in the agitator lines or cones that cannot be removed with vacuum. Be sure the area is clear, and you are not in the line of fire when you open it.

6.14. Diesel Flush the Hibon for 1 to 2 seconds on vacuum. Diesel flushing is a very good way of keeping your system from freezing up in the winter, and keeping the whole system lubricated so nothing will stick to the piping or the pump itself.

6.15. Disengage Hibon, remove spread plate, install caps, ensure 4-way valve is on vacuum, re-enter truck, use clutch when disengaging the main PTO and Agitator PTO, place transmission in proper gear and leave the spread field. If using the Tire Boss System then inflate tires. **Important, Vacuum Truck Must Leave the Spread Field on Vacuum or Neutral.**

6.16. Before you leave the spread field, ensure your spread is straight and the proper length with no pooling or mud puddles left behind. Spills and shale piles must be cleaned up and reported to your tech.

6.16.1. **Automatic:** once the spread plate is off and the caps are back on, don't forget to put the transfer case back into high range. If you are spreading right off location you might want to leave the transfer case in low range the whole time, or until you need to go for fuel.

6.17. Ensure before you drive back onto location that the 4-way valve is on vacuum.

REPORT ANY PROBLEMS TO YOUR SUPERVISOR OR TECHICIAN!

Remember: if you are in doubt about what to do, call your supervisor